

Nonparametric Statistics

Introduction and the Empirical Distribution Function

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What is Nonparametric Statistics?

Parametric Approach

The model is fully specified by a **finite** number of parameters $\theta \in \Theta \subseteq \mathbb{R}^d$:

$$X \sim P_\theta, \quad d < \infty$$

Examples: $\mathcal{N}(\mu, \sigma^2)$, $\text{Poisson}(\lambda)$, $\text{Bernoulli}(p)$

Nonparametric Approach

The model lives in an **infinite-dimensional** space — minimal assumptions on F :

$$X \sim F, \quad F \in \mathcal{F}$$

Examples: any continuous distribution, density estimation, flexible regression

Idea (Wasserman, 2006): *“The basic idea of nonparametric inference is to use data to infer an unknown quantity while making as few assumptions as possible.”*

Advantages and Disadvantages

Advantages

- ✓ **Flexibility:** models complex structures without imposing a functional form
- ✓ **Robustness:** less sensitive to outliers and to model mis-specification
- ✓ **Generality:** works with ordinal data (rank-based methods)
- ✓ **Validity:** protects against parametric mis-specification risk

Disadvantages

- ✗ **Efficiency:** if the parametric model is correct, parametric methods are more efficient (though the loss is often small)
- ✗ **Interpretation:** sometimes harder to interpret results
- ✗ **Computation:** can be more costly (mitigated by modern computing)
- ✗ **Sample size:** often requires more data

Fifty years ago most nonparametric methods were not feasible due to limited computing power. Rapid developments in computing science have changed the situation entirely.

Fundamental Problems in Nonparametric Inference

1. Estimating the distribution function

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F \Rightarrow$ estimate $F(x) = \mathbb{P}(X \leq x)$

(this lecture)

2. Estimating statistical functionals

Estimate $T(F)$, e.g. the mean $T(F) = \int x dF(x)$

3. Density estimation

Estimate $f(x) = F'(x)$ from $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$

4. Nonparametric regression

Given $(X_1, Y_1), \dots, (X_n, Y_n)$, estimate $r(x) = \mathbb{E}[Y|X = x]$

5. Hypothesis testing

Goodness-of-fit tests, two-sample tests, rank tests

In all problems, the distribution F (or density f , or regression r) is assumed to lie in a large, possibly infinite-dimensional model class $\mathcal{F}$.

The Cumulative Distribution Function

Definition

For a random variable X , the **cumulative distribution function (CDF)** is

$$F(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

Key properties:

- (Range) $0 \leq F(x) \leq 1$
- (Monotonicity) $x \leq y \Rightarrow F(x) \leq F(y)$
- (Right-continuity) $\lim_{x \rightarrow y^+} F(x) = F(y)$
- $\lim_{x \rightarrow -\infty} F(x) = 0$
- $\lim_{x \rightarrow +\infty} F(x) = 1$
- $\mathbb{P}(X = x) = F(x) - F(x^-)$

Why CDF rather than PDF/PMF?

The CDF **always exists** — even when X is neither continuous nor discrete. The PDF exists only for continuous r.v., the PMF only for discrete r.v. The CDF is the universal tool.

Classical CDF Examples

Uniform $\mathcal{U}([0, 1])$

$$F(x) = \begin{cases} 0 & x < 0 \\ x & x \in [0, 1] \\ 1 & x > 1 \end{cases}$$



Exponential $\text{Exp}(\lambda)$, $\lambda = 0.5$

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$



A Mixed Random Variable (no PDF, no PMF)

Let $X = 0.5$ with probability $1/2$; otherwise $X \sim \mathcal{U}([0, 1])$. Then X has neither a PDF nor a PMF, but its CDF exists. $\Rightarrow$ The CDF is the *only* unified representation.

Generalized Inverse (Quantile Function)

$$F^{-1}(u) = \inf\{x : F(x) \geq u\}, \quad u \in (0, 1).$$

Theorem (Inversion Method)

Let $U \sim \mathcal{U}([0, 1])$. Then $W = F^{-1}(U)$ has CDF equal to F .

Conversely, if $X \sim F$ is continuous, then $V = F(X) \sim \mathcal{U}([0, 1])$.

Application: Simulating $\text{Exp}(\lambda)$ from a Uniform

Since $F(x) = 1 - e^{-\lambda x}$, we have $F^{-1}(u) = -\frac{1}{\lambda} \log(1 - u)$.

Algorithm: Generate $U \sim \mathcal{U}([0, 1])$, set $X = -\frac{1}{\lambda} \log(1 - U) \sim \text{Exp}(\lambda)$.

Practical importance: The inversion method underlies most random number generators and is the backbone of Monte Carlo simulation.

Motivation: Estimating the CDF from Data

Problem: In practice, F is **unknown**. All we have is a sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$.

Natural reasoning:

For any fixed x_0 ,

$$F(x_0) = \mathbb{P}(X_i \leq x_0)$$

is a *probability*, naturally estimated by a *frequency*:

$$\hat{F}_n(x_0) = \frac{\#\{X_i \leq x_0\}}{n} = \frac{1}{n} \sum_{i=1}^n 1(X_i \leq x_0).$$

Key insight: The Law of Large Numbers guarantees that a frequency converges to its probability.

Applying this *for every* x simultaneously yields an estimator of the entire function F .

Question:

Given a random sample $X_1, \dots, X_n$, how can we best estimate the unknown CDF F ?

Definition of the Empirical Distribution Function

Definition — Empirical Distribution Function (EDF / ECDF)

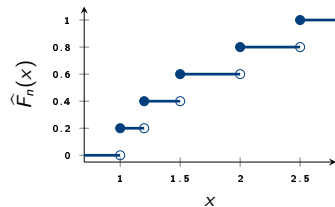
Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$. The **empirical distribution function** is defined as

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n 1(X_i \leq x), \quad x \in \mathbb{R}.$$

It is the CDF that places mass $1/n$ at each observation X_i .

Structure of ECDF

- A *step function* with exactly n jumps
- Each jump occurs at an order statistic $X_{(i)}$
- Each jump has height $\frac{1}{n}$
- $\hat{F}_n(x) \in \{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}$

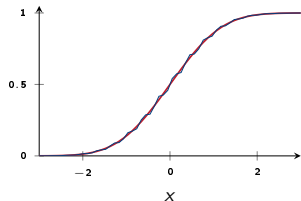


EDF of $\{1, 1.2, 1.5, 2, 2.5\}$

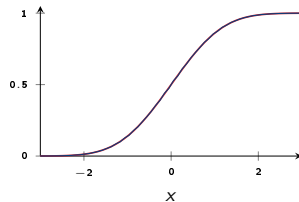
Visual Convergence: ECDF $\rightarrow$ CDF

As n grows, the EDF approaches the true CDF. Here the true CDF is Φ , the standard normal $\mathcal{N}(0,1)$ CDF.

$n = 100$: **fluctuations still visible**



$n = 1000$: **EDF and CDF nearly identical**



— True CDF $\Phi(x)$

— Empirical CDF $\hat{F}_n(x)$

Connection to the Bernoulli Distribution

For fixed x , define $Y_i = 1(X_i \leq x)$.

Y_i is a Bernoulli random variable!

$$Y_i = \begin{cases} 1 & \text{if } X_i \leq x \\ 0 & \text{if } X_i > x \end{cases}$$

Parameter: $p = \mathbb{P}(Y_i = 1) = \mathbb{P}(X_i \leq x) = F(x)$

$$\Rightarrow Y_i \sim B(F(x))$$

$$\mathbb{E}[Y_i] = F(x), \quad \mathbb{V}[Y_i] = F(x)(1 - F(x))$$

Since $\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n Y_i$:

$$n\hat{F}_n(x) \sim B(n, F(x))$$

Mean and Variance of $\hat{F}_n(x)$

For any fixed $x \in \mathbb{R}$:

$$\mathbb{E}[\hat{F}_n(x)] = F(x), \quad \mathbb{V}[\hat{F}_n(x)] = \frac{F(x)(1 - F(x))}{n}.$$

Unbiasedness, Consistency, Asymptotic Normality

Theorem — Fundamental Statistical Properties of EDF

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$. For any fixed $x \in \mathbb{R}$:

- ➊ **Unbiasedness:** $\text{Bias}[\hat{F}_n(x)] = \mathbb{E}[\hat{F}_n(x)] - F(x) = 0$
- ➋ **Consistency:** $\mathbb{V}[\hat{F}_n(x)] = \frac{F(x)(1 - F(x))}{n} \rightarrow 0$ as $n \rightarrow \infty$, so $\hat{F}_n(x) \xrightarrow{P} F(x)$
- ➌ **Asymptotic Normality (CLT):**

$$\sqrt{n}(\hat{F}_n(x) - F(x)) \xrightarrow{D} \mathcal{N}(0, F(x)(1 - F(x)))$$

Numerical Example

$X_1, \dots, X_{100} \stackrel{\text{iid}}{\sim} \mathcal{U}([0, 2])$. Find the distribution of $\hat{F}_n(0.8)$.

$$F(0.8) = 0.8/2 = 0.4, \text{ so: } \mathbb{E}[\hat{F}_n(0.8)] = 0.4 \quad \text{and} \quad \mathbb{V}[\hat{F}_n(0.8)] = \frac{0.4 \times 0.6}{100} = 2.4 \times 10^{-3}.$$

The Glivenko–Cantelli Theorem

The previous properties are *pointwise* (for a fixed x). We can do much better:

Glivenko–Cantelli Theorem — Uniform Convergence

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

The ECDF converges to F **almost surely** and **uniformly** over all $x \in \mathbb{R}$.

The Glivenko–Cantelli Theorem

Why this matters:

- Uniform convergence is *much stronger* than pointwise
- It guarantees $\hat{F}_n$ approximates the *entire* CDF simultaneously
- It is the theoretical foundation of the **bootstrap** and resampling methods
- It justifies using $\hat{F}_n$ as a substitute for F in any continuous functional

Plug-in Principle:

For any functional T continuous with respect to the sup-norm:

$$T(\hat{F}_n) \xrightarrow{a.s.} T(F).$$

Estimating $T(F)$ by $T(\hat{F}_n)$ is **consistent**.

The DKW Inequality and Confidence Bands

Dvoretzky–Kiefer–Wolfowitz (DKW) Inequality

For any $\varepsilon > 0$ and any $n \geq 1$:

$$\mathbb{P}\left(\sup_{x \in \mathbb{R}} |F(x) - \hat{F}_n(x)| > \varepsilon\right) \leq 2 e^{-2n\varepsilon^2}.$$

Constructing a confidence band

Set $\varepsilon_n = \sqrt{\frac{\log(2/\alpha)}{2n}}$ and define

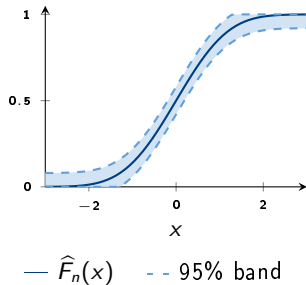
$$L(x) = \max\{\hat{F}_n(x) - \varepsilon_n, 0\}$$

$$U(x) = \min\{\hat{F}_n(x) + \varepsilon_n, 1\}$$

Then, for **any** F and any n :

$$\mathbb{P}(L(x) \leq F(x) \leq U(x) \forall x) \geq 1 - \alpha.$$

This is a **finite-sample** (non-asymptotic) guarantee.



Confidence Interval for $F(x)$ at a Fixed Point

Setting: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$. Fix a point x_0 . We want a confidence interval for $F(x_0) = \mathbb{P}(X \leq x_0)$.

Define $Y_i = 1(X_i \leq x_0)$. Then $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p)$ with $p = \mathbb{P}(X_i \leq x_0) = F(x_0)$.
The EDF gives the natural estimator:

$$\hat{p}_n = \hat{F}_n(x_0) = \frac{1}{n} \sum_{i=1}^n Y_i = \frac{\#\{X_i \leq x_0\}}{n}.$$

Moreover: $n \hat{p}_n \sim B(n, p)$.

A CI for $F(x_0)$ is exactly a CI for a binomial proportion p .

Wald Confidence Interval (Asymptotic)

Wald Confidence Interval

$$\hat{p}_n \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_n(1 - \hat{p}_n)}{n}}$$

Derivation:

By the Central Limit Theorem, for large n :

$$\frac{\hat{p}_n - p}{\sqrt{p(1 - p)/n}} \xrightarrow{D} \mathcal{N}(0, 1).$$

Replace unknown p by $\hat{p}_n$ in the variance (Slutsky):

$$\frac{\hat{p}_n - p}{\sqrt{\hat{p}_n(1 - \hat{p}_n)/n}} \xrightarrow{D} \mathcal{N}(0, 1).$$

Inverting the two-sided test gives the Wald CI.

Numerical Example

$n = 100$, $x_0 = 0.8$, $X_i \stackrel{\text{iid}}{\sim} \mathcal{U}([0, 2])$.

$$\hat{p}_n = \frac{F(0.8)}{\sqrt{0.4 \times 0.6/100}} = 0.4, \quad \widehat{\text{se}} = 0.049.$$

95% Wald CI:

$$0.4 \pm 1.96 \times 0.049 = (0.304, 0.496).$$

Wald CI: Properties and Limitations

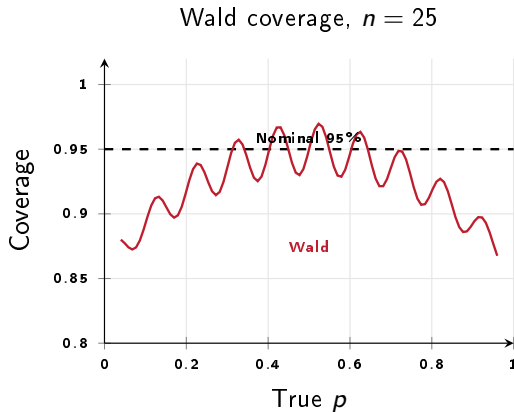
Advantages

- ✓ Extremely **simple** to compute
- ✓ Intuitive: center $\pm$ margin of error
- ✓ Good accuracy when n is large and p is away from 0 and 1
- ✓ Universal: does not require any special tables

Disadvantages

- ✗ **Poor coverage** near $p = 0$ or $p = 1$: the normal approximation fails
- ✗ Can give **impossible bounds**: lower limit < 0 or upper limit > 1
- ✗ Actual coverage can be much less than nominal $(1 - \alpha)$ for small n
- ✗ Tends to **under-cover** (coverage $< 1 - \alpha$)

Wald CI: Properties and Limitations



Rule of thumb: The Wald CI is acceptable only when $n\hat{p}_n \geq 5$ and $n(1 - \hat{p}_n) \geq 5$.

Motivation: The Wald CI is asymptotic. Can we get a **finite-sample** guarantee, valid for any n ?

Hoeffding's Inequality

Let $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim}$ with $Y_i \in [0, 1]$ and $\mu = \mathbb{E}[Y_i]$. Then for any $\varepsilon > 0$:

$$\mathbb{P}(|\bar{Y}_n - \mu| > \varepsilon) \leq 2 e^{-2n\varepsilon^2}.$$

Applying Hoeffding to our setting:

- We have $Y_i = 1(X_i \leq x_0) \in \{0, 1\}$, $\bar{Y}_n = \hat{p}_n$, $\mu = p = F(x_0)$.
- Set the bound equal to α :

$$2e^{-2n\varepsilon^2} = \alpha \Rightarrow \varepsilon_n = \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

- Then: $\mathbb{P}(|\hat{p}_n - p| > \varepsilon_n) \leq \alpha$.

Hoeffding Confidence Interval

$$\hat{p}_n \pm \varepsilon_n, \quad \text{where} \quad \varepsilon_n = \sqrt{\frac{\log(2/\alpha)}{2n}}.$$

Equivalently: $\hat{p}_n - \varepsilon_n \leq p \leq \hat{p}_n + \varepsilon_n$ holds with probability $\geq 1 - \alpha$.

Hoeffding: Numerical example

Example

$n = 100$, $\hat{p}_n = 0.4$, $\alpha = 0.05$.

$$\varepsilon_n = \sqrt{\log(40)/(200)} = \sqrt{3.689/200} = \sqrt{0.0184} = 0.136.$$

Hoeffding CI: (0.264, 0.536)

Compare:

Wald: (0.304, 0.496) — narrower

Hoeffding is **wider** — the price of a finite-sample guarantee.

Half-width: $\varepsilon_n = \sqrt{\log(2/\alpha)/(2n)}$

For $\alpha = 0.05$: $\varepsilon_n \approx 1.36/\sqrt{n}$

For $\alpha = 0.01$: $\varepsilon_n \approx 1.63/\sqrt{n}$

Hoeffding: Connection to DKW

Connection to DKW Inequality

The DKW inequality gives a **simultaneous** (uniform) bound:

$$\mathbb{P}\left(\sup_{x \in \mathbb{R}} |F(x) - \hat{F}_n(x)| > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

With $\varepsilon_n = \sqrt{\log(2/\alpha)/(2n)}$:

$$\mathbb{P}(F(x) \in [\hat{F}_n(x) - \varepsilon_n, \hat{F}_n(x) + \varepsilon_n] \text{ for all } x) \geq 1 - \alpha.$$

The Hoeffding CI at a **single point** x_0 follows immediately from this uniform bound (which is actually stronger).

When to use Hoeffding

Use when a **formal finite-sample guarantee** is needed, e.g. in hypothesis testing with multiple comparisons or when n is very small. Acceptable for the price of wider intervals.

What is a Statistical Functional?

Definition — Statistical Functional

A **statistical functional** is any mapping $T : \mathcal{F} \rightarrow \mathbb{R}$ (or to a function space) that assigns a real-valued quantity to each CDF F .

$$T(F) \in \mathbb{R}.$$

Idea (Wasserman, 2006): Rather than making parametric assumptions, we express every quantity of interest as a functional of the CDF F . Then we *estimate* the CDF by $\hat{F}_n$ and *plug it in*.

This strategy is completely general: it works for the mean, the variance, quantiles, correlations, skewness, tail probabilities — and more.

Linear Functionals

Linear Functional

A functional of the form

$$T(F) = \int a(x) dF(x)$$

is called a **linear functional**. The function $a(\cdot)$ is fixed.

Examples of Linear Functionals

Quantity	Function $a(x)$	Functional $T(F)$
Mean	$a(x) = x$	$\int x dF(x) = \mu$
k -th moment	$a(x) = x^k$	$\int x^k dF(x) = \mathbb{E}[X^k]$
Tail probability	$a(x) = 1(x > c)$	$\mathbb{P}(X > c)$
CDF value	$a(x) = 1(x \leq t)$	$F(t)$

Non-Linear Functionals

Non-Linear Functionals

Many quantities of interest are **not** of the form $\int a dF$. They are *non-linear* in F , often written as $a(T_1(F), \dots, T_k(F))$.

Examples of Non-Linear Functionals

Quantity	Functional $T(F)$
Variance	$\int (x - \mu)^2 dF(x) = \int x^2 dF - \left(\int x dF\right)^2$
Skewness	$\frac{\int (x - \mu)^3 dF}{\left(\int (x - \mu)^2 dF\right)^{3/2}}$
Median	$F^{-1}(1/2)$
Correlation (bivariate)	$\frac{\int (x - \mu_X)(y - \mu_Y) dF}{\sigma_X \sigma_Y}$

The Plug-in Principle

Plug-in Estimator (Wasserman, 2006)

The **plug-in estimator** of $\theta = T(F)$ is

$$\hat{\theta}_n = T(\hat{F}_n),$$

obtained by replacing the unknown CDF F by the empirical CDF $\hat{F}_n$.

Why It Works: Glivenko–Cantelli

By the Glivenko–Cantelli theorem, $\|\hat{F}_n - F\|_\infty \xrightarrow{a.s.} 0$.

If T is **continuous** with respect to the sup-norm $\|F - G\|_\infty = \sup_x |F(x) - G(x)|$, then:

$$T(\hat{F}_n) \xrightarrow{a.s.} T(F).$$

The **plug-in estimator** is always consistent.

The Empirical Measure $d\hat{F}_n$

The Empirical Measure

The EDF $\hat{F}_n$ induces a discrete **probability measure** on $\mathbb{R}$, called the **empirical measure**, which assigns mass $\frac{1}{n}$ to each observation:

$$d\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}(x),$$

where δ_{X_i} is the **Dirac mass** at X_i (i.e. a unit point mass concentrated at X_i).

For any measurable set A :

$$\int_A d\hat{F}_n(x) = \hat{F}_n(A) = \frac{\#\{X_i \in A\}}{n}.$$

The Empirical Measure $d\hat{F}_n$

Riemann–Stieltjes integral against $d\hat{F}_n$: Since $\hat{F}_n$ is a step function that jumps by $\frac{1}{n}$ at each X_i , integrating any function a against $d\hat{F}_n$ reduces to a finite sum:

$$\int_{\mathbb{R}} a(x) d\hat{F}_n(x) = \sum_{i=1}^n a(X_i) \cdot \underbrace{\Delta \hat{F}_n(X_i)}_{=1/n} = \frac{1}{n} \sum_{i=1}^n a(X_i).$$

This identity is the key to computing **all** plug-in estimators of linear functionals.

Plug-in for Linear Functionals

Formula for Linear Functionals

If $T(F) = \int a(x) dF(x)$, then the plug-in estimator has a closed form. Since $\hat{F}_n$ places mass $1/n$ at each X_i :

$$T(\hat{F}_n) = \int a(x) d\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n a(X_i).$$

Plug-in for the Mean and Variance

Mean: $T(F) = \int x dF(x)$.

$$T(\hat{F}_n) = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n.$$

Variance: $T(F) = \int x^2 dF - (\int x dF)^2$.

$$T(\hat{F}_n) = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Plug-in for Non-Linear Functionals

Plug-in for the Skewness

$T(F) = \frac{\int (x - \mu)^3 dF}{(\int (x - \mu)^2 dF)^{3/2}}$. Replace F by $\hat{F}_n$, using $\mu \rightarrow \bar{X}_n$:

$$T(\hat{F}_n) = \hat{\kappa}_n = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^3}{(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2)^{3/2}}.$$

Plug-in for the Sample Quantile

$T(F) = F^{-1}(p) = \inf\{x : F(x) \geq p\}$ is the p -th quantile.

The plug-in estimator is the **sample quantile**:

$$T(\hat{F}_n) = \hat{F}_n^{-1}(p) = \inf\{x : \hat{F}_n(x) \geq p\} = X_{(\lceil np \rceil)}.$$

At all continuity points of F^{-1} : $T(\hat{F}_n) \xrightarrow{a.s.} T(F)$.

Goodness-of-Fit Testing

Problem Statement

Given $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ (unknown), test:

$$H_0 : F = F_0 \quad \text{vs.} \quad H_1 : F \neq F_0,$$

where F_0 is a **fully specified** distribution (e.g. $\mathcal{N}(0, 1)$, $\text{Exp}(1)$, etc.).

Common principle: Under H_0 , the empirical CDF $\hat{F}_n$ should be close to F_0 . By Glivenko–Cantelli, $\hat{F}_n \xrightarrow{a.s.} F$. If H_0 is true, then $\hat{F}_n \xrightarrow{a.s.} F_0$.

The three classical tests differ in *how they measure the distance* between $\hat{F}_n$ and F_0 :

- KS: **maximum deviation** (sup-norm)
- Cramér–von Mises: **average squared deviation** (L^2 -norm)
- Anderson–Darling: **weighted L^2 -norm** (emphasizing tails)

Test 1 — Kolmogorov–Smirnov (KS)

KS Test Statistic

$$T_{\text{KS}} = \sqrt{n} \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F_0(x)|.$$

Measures the **maximum pointwise deviation** between $\hat{F}_n$ and F_0 .

Asymptotic Distribution Under H_0 (Kolmogorov)

Under H_0 , if F_0 is continuous:

$$T_{\text{KS}} \xrightarrow{D} K, \quad \text{where} \quad \mathbb{P}(K \leq t) = 1 - 2 \sum_{j=1}^{\infty} (-1)^{j+1} e^{-2j^2 t^2}.$$

This is the **Kolmogorov distribution**, tabulated and distribution-free.

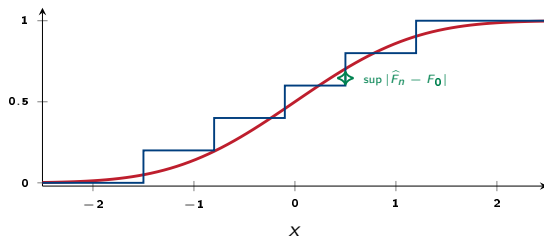
Critical values at level α : $t_{0.10} \approx 1.22$, $t_{0.05} \approx 1.36$, $t_{0.01} \approx 1.63$.

KS Test: Computation and Visual Interpretation

How to Compute T_{KS} in Practice

Since $\hat{F}_n$ is a step function with jumps at the order statistics $X_{(1)} \leq \dots \leq X_{(n)}$, the supremum is attained at one of these points:

$$T_{KS} = \sqrt{n} \max_{1 \leq i \leq n} \max \left(\left| \frac{i}{n} - F_0(X_{(i)}) \right|, \left| \frac{i-1}{n} - F_0(X_{(i)}) \right| \right).$$



Test 2 — Cramér–von Mises

Cramér–von Mises Test Statistic

$$T_{\text{CM}} = n \int_{-\infty}^{+\infty} [\hat{F}_n(x) - F_0(x)]^2 dF_0(x).$$

Measures the **average squared deviation** between $\hat{F}_n$ and F_0 , weighted by F_0 (i.e. the L^2 distance with respect to F_0).

Computational Formula and Asymptotic Distribution

Using the order statistics $X_{(1)} \leq \dots \leq X_{(n)}$:

$$T_{\text{CM}} = \frac{1}{12n} + \sum_{i=1}^n \left(F_0(X_{(i)}) - \frac{2i-1}{2n} \right)^2.$$

Under H_0 , $T_{\text{CM}} \xrightarrow{D} \int_0^1 [B(t)]^2 dt$, where B is a **Brownian bridge**. The limiting distribution does not depend on F_0 (distribution-free).

Test 3 — Anderson–Darling

Anderson–Darling Test Statistic

$$T_{\text{AD}} = n \int_{-\infty}^{+\infty} \frac{[\hat{F}_n(x) - F_0(x)]^2}{F_0(x)[1 - F_0(x)]} dF_0(x).$$

A **weighted** version of the CvM statistic: the weight $1/[F_0(x)(1 - F_0(x))]$ diverges near $x = \pm\infty$, giving **extra sensitivity to tail deviations**.

Computational Formula

$$T_{\text{AD}} = -n - \sum_{i=1}^n \frac{2i-1}{n} [\ln F_0(X_{(i)}) + \ln(1 - F_0(X_{(n+1-i)}))].$$

Under H_0 , $T_{\text{AD}} \xrightarrow{D} \int_0^1 \frac{[B(t)]^2}{t(1-t)} dt$ (weighted Brownian bridge). Critical values depend on the specific F_0 being tested.

Comparison: KS, CvM, and Anderson–Darling

	KS	CvM	AD
Distance	L_∞	L^2	Weighted L^2
Tail sensitivity	Low	Medium	High
Distribution-free	Yes	Yes	Approx.
Detects	Max deviation	Global deviation	Tail deviation

Practical advice: CvM and AD are generally **more powerful** than KS because they use information at all points, not just the maximum. Use AD when tail behavior is of primary concern.

Two-Sample Testing: Setup

Problem Statement

Given two independent samples: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F_X$ and $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} F_Y$.

Test:

$$H_0 : F_X = F_Y \quad \text{vs.} \quad H_1 : F_X \neq F_Y.$$

No parametric form is assumed for F_X or F_Y .

Strategy: Estimate F_X and F_Y by their EDFs $\hat{F}_X$ and $\hat{F}_Y$. Under H_0 , both EDFs should converge to the same limit, so they should be close to each other.

The two-sample tests directly mirror the one-sample tests, replacing $\hat{F}_n - F_0$ by $\hat{F}_X - \hat{F}_Y$.

Define the **pooled EDF**: $\hat{H}(t) = \frac{n\hat{F}_X(t) + m\hat{F}_Y(t)}{n + m}$.

Two-Sample KS Test

Two-Sample KS Statistic (Smirnov Test)

$$T_{\text{KS}} = \sqrt{\frac{nm}{n+m}} \sup_{x \in \mathbb{R}} |\hat{F}_X(x) - \hat{F}_Y(x)|.$$

Asymptotic Distribution Under H_0 (Smirnov)

Under H_0 (and as $n, m \rightarrow \infty$ with $n/m \rightarrow \lambda \in (0, \infty)$):

$$T_{\text{KS}} \xrightarrow{D} K \quad (\text{Kolmogorov distribution}).$$

The p -value and critical values are the **same** as in the one-sample KS test.

Two-Sample Cramér–von Mises and Anderson–Darling

Two-Sample CvM Statistic

$$T_{\text{CM}} = \frac{nm}{n+m} \int_{-\infty}^{+\infty} [\hat{F}_X(x) - \hat{F}_Y(x)]^2 d\hat{H}(x).$$

Equivalent to summing squared differences at all combined order statistics.

Two-Sample Anderson–Darling Statistic

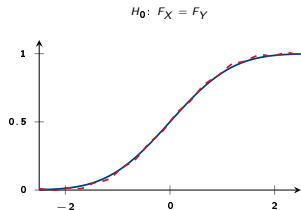
$$T_{\text{AD}} = \frac{nm}{n+m} \int_{-\infty}^{+\infty} \frac{[\hat{F}_X(x) - \hat{F}_Y(x)]^2}{\hat{H}(x)[1 - \hat{H}(x)]} d\hat{H}(x).$$

Upweights tail regions using the pooled EDF $\hat{H}$ as the reference.

Two-Sample Tests: Visual Illustration

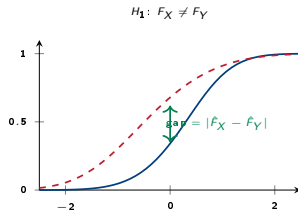
Under H_0 ($F_X = F_Y$):

both EDFs converge to the same CDF; they should be uniformly close.



Under H_1 ($F_X \neq F_Y$):

the two EDFs diverge; the sup (or integral) of their difference is large.



One-Sample vs. Two-Sample: Unified View

Test	One-sample	Two-sample
KS	$\sqrt{n} \sup_x \hat{F}_n - F_0 $	$\sqrt{\frac{nm}{n+m}} \sup_x \hat{F}_X - \hat{F}_Y $
CvM	$n \int (\hat{F}_n - F_0)^2 dF_0$	$\frac{nm}{n+m} \int (\hat{F}_X - \hat{F}_Y)^2 d\hat{H}$
AD	$n \int \frac{(\hat{F}_n - F_0)^2}{F_0(1-F_0)} dF_0$	$\frac{nm}{n+m} \int \frac{(\hat{F}_X - \hat{F}_Y)^2}{\hat{H}(1-\hat{H})} d\hat{H}$
Reference	Known F_0	Pooled EDF $\hat{H}$

Unifying principle: All six test statistics measure how far the empirical CDF(s) deviate from the expected CDF under H_0 . In the two-sample case, F_0 is *unknown* and is estimated by the pooled EDF $\hat{H}$. The normalization $nm/(n+m)$ replaces n .